Task 4
Group Activity
Using python in-built library Numpy:

Solar Power Generation & Maintenance Advisor can be developed using NumPy. NumPy is useful for numerical calculations and analyzing solar generation data.

- Descriptive analytics: Calculate average, maximum, minimum, and total energy generation using NumPy.
- Diagnostic analytics: Calculate the difference between expected and actual energy output.
- Predictive analytics: Prepare numerical data for machine learning models. Scikit-learn can use NumPy arrays.
- Prescriptive analytics: Calculate generation losses and estimate the benefits of maintenance actions.

```
import numpy as np

expected = np.array([45, 50, 45, 40, 48])
actual = np.array([42, 48, 35, 30, 45])

shortfall = expected - actual

print("Average generation:", np.mean(actual))
print("Total generation:", np.sum(actual))
print("Generation shortfall:", shortfall)
```

Without using the python in-built library Numpy

The Solar Power Generation & Maintenance Advisor can be developed without directly using NumPy. Python and Pandas can be used to clean data, calculate total and average energy generation, and compare expected output with actual output. Diagnostic analytics helps identify possible reasons for low generation, while Scikit-learn is used to predict future solar energy production. Prescriptive analytics uses Python conditions and calculations to recommend maintenance actions, such as panel cleaning or inverter inspection. Matplotlib is used to visualize generation trends, Streamlit creates an interactive dashboard, and MySQL stores historical data. Although NumPy is not required in your own code, Scikit-learn and other libraries may use it internally.

```
import pandas as pd

data = pd.DataFrame({
    "Expected": [45, 50, 45, 40, 48],
    "Actual": [42, 48, 35, 30, 45]
})
```

```
data["Shortfall"] = data["Expected"] - data["Actual"]

print("Average generation:", data["Actual"].mean())
print("Total generation:", data["Actual"].sum())
print(data)
```